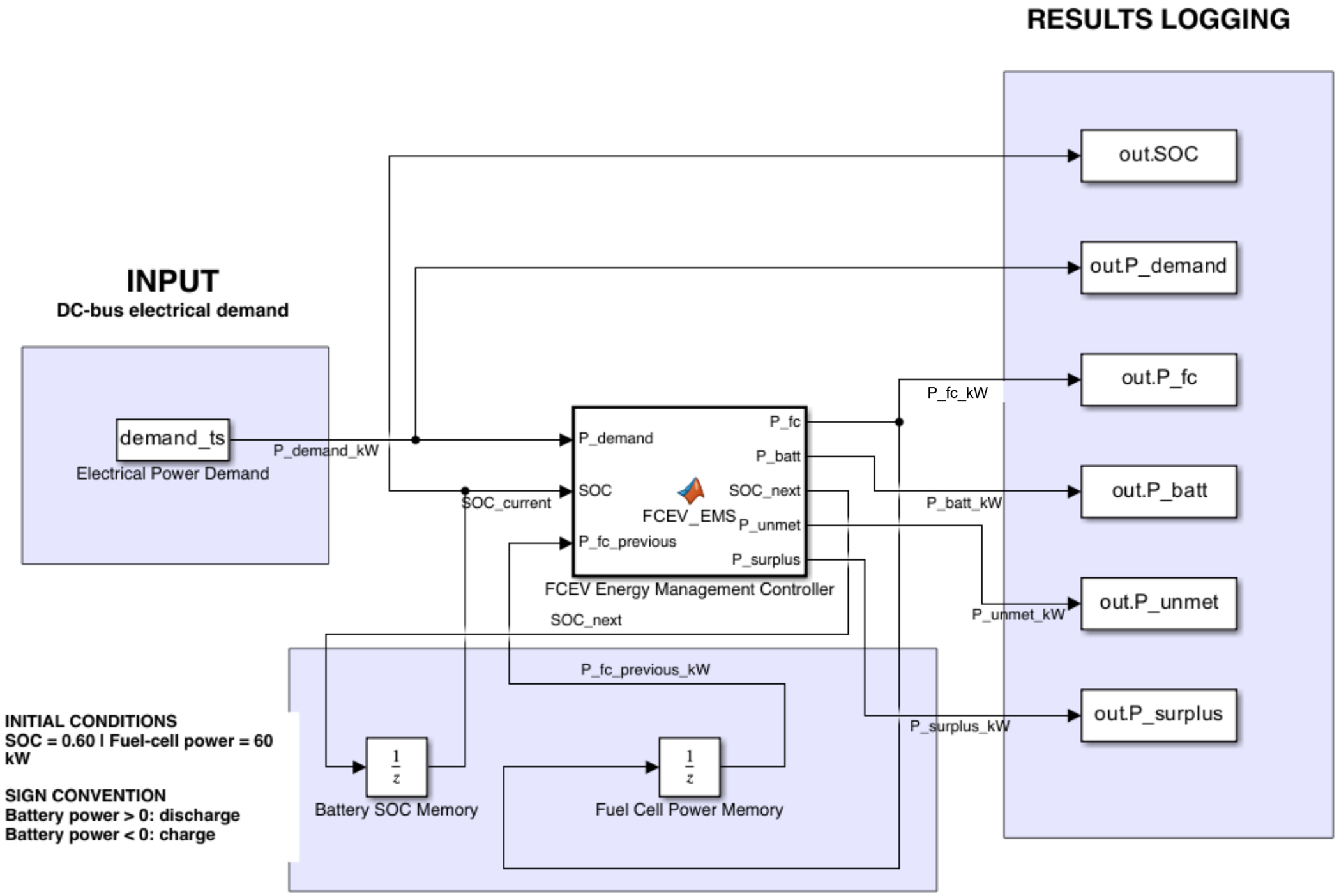
Hybrid FCEV power sharing

18-tonne vehicle | 120 kW fuel cell | 30 kWh battery | Sample time: 0.1 s



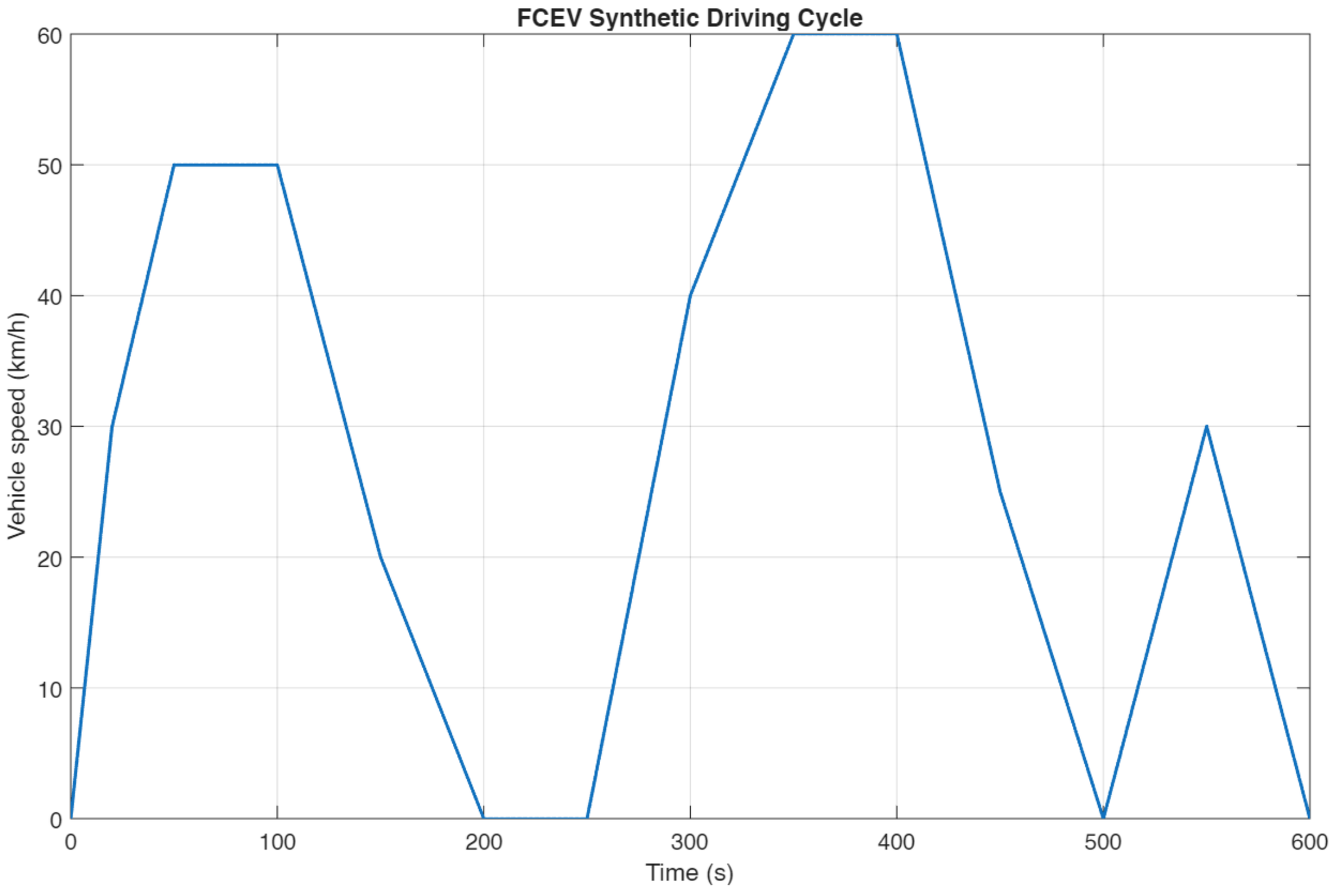
CONTROL CONCEPT

The fuel cell changes its power gradually. The battery supplies extra power when needed and stores energy from braking or excess fuel-cell output. The controller also checks the battery's charge level.

Scope: power-based simulation with a synthetic driving cycle.

Start with the driving task

A prescribed speed trace defines acceleration, cruising, braking and stops.

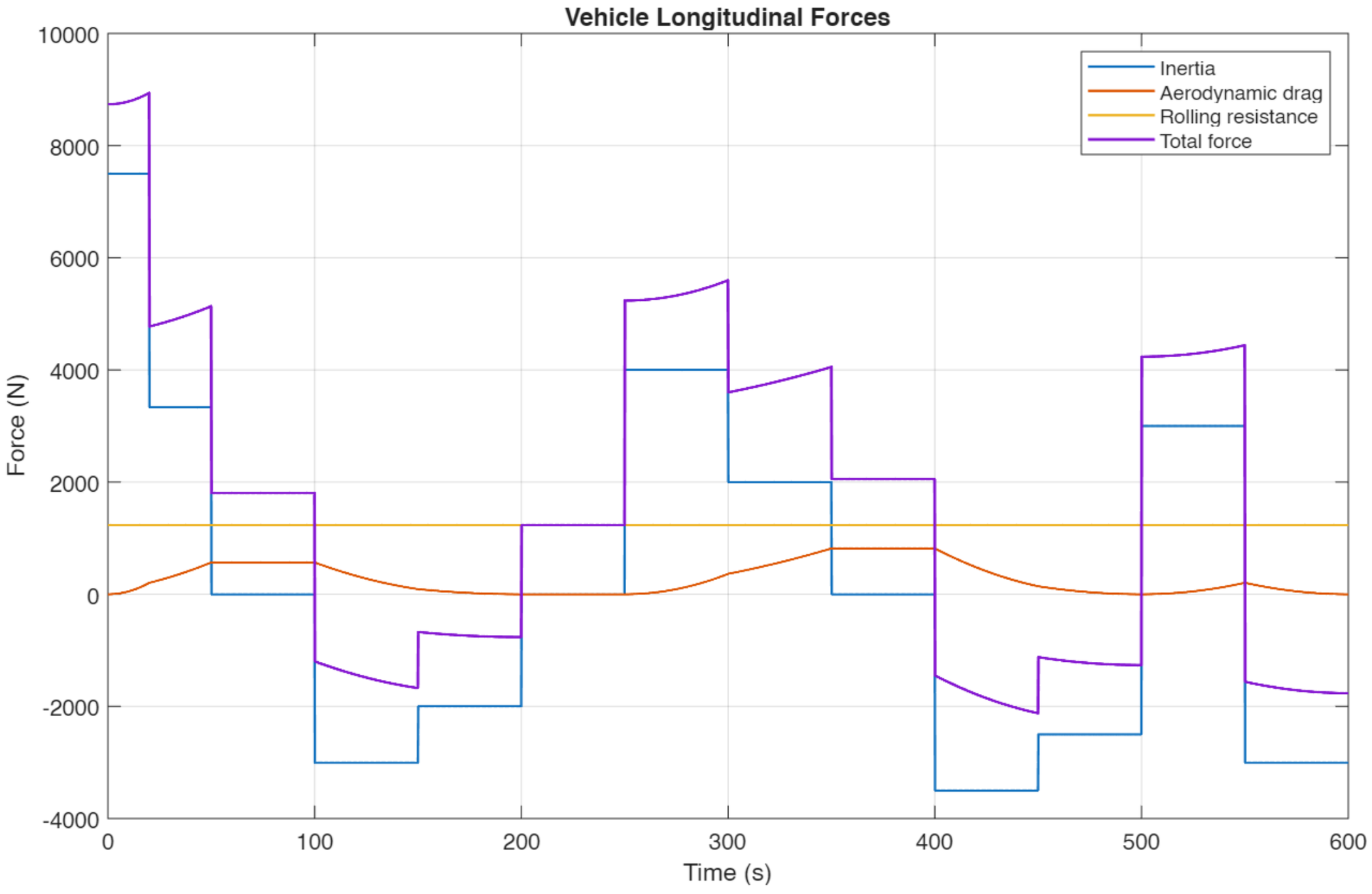


600 SECONDS | 4.72 KM | 60 KM/H MAXIMUM

Upward slopes mean acceleration; downward slopes mean deceleration. This trace is synthetic, not measured vehicle data.

Translate speed into force

Wheel force combines inertia, aerodynamic drag and rolling resistance on a flat road.

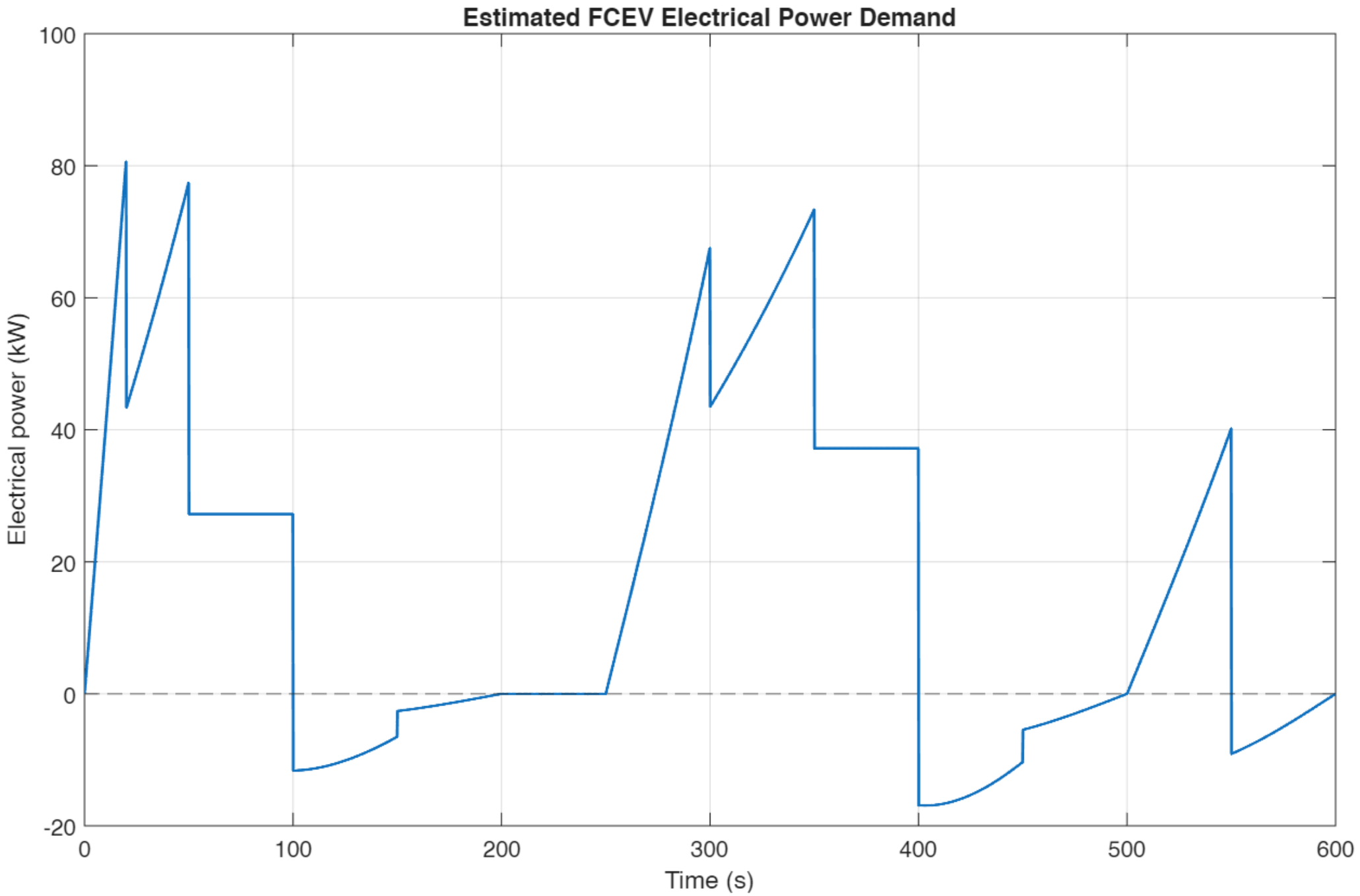


F_total = m a + 0.5 rho Cd A v^2 + Crr m g

Acceleration dominates the largest force peaks. At constant speed, traction still overcomes drag and rolling resistance.

Translate force into power

Wheel power is force multiplied by speed; drivetrain efficiencies convert it to electrical demand.

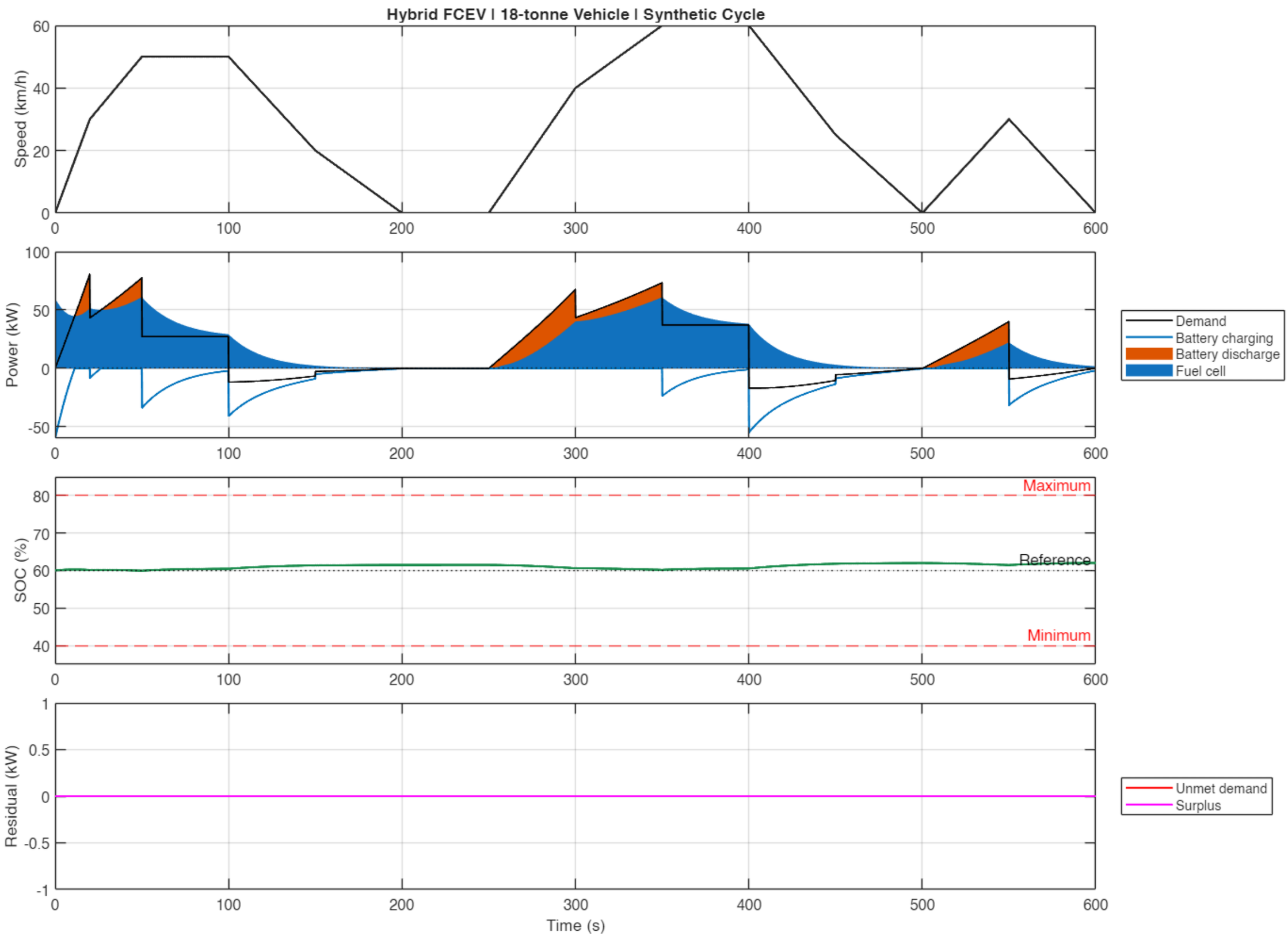


POSITIVE: TRACTION | NEGATIVE: AVAILABLE REGENERATION

Acceleration peaks reach about 80 kW. Negative demand represents recoverable braking power before battery acceptance limits.

Let the battery handle transients

Fuel-cell smoothing and battery power limits determine the split at each 0.1-second sample.



Battery charging can include both braking energy and fuel-cell output. The plotted residuals are approximately zero.

Demand met. SOC stayed in range.

Results from one 600-second synthetic cycle; component limits and efficiencies are model assumptions.

SOC range	**59.95-62.03%**
Final SOC	**62.03% (initial: 60.00%)**
Fuel-cell electrical energy	**3.384 kWh**
Unmet-demand energy	**0.000000 kWh**
Surplus energy	**Approximately zero**
Maximum power-balance error	**3.553 x 10^-15 kW**

WHAT THIS RESULT MEANS

The battery gained about 0.609 kWh of stored energy. This run is net charging; it does not establish hydrogen efficiency or long-term SOC stability.

NEXT: TEST AN AUTHENTIC TRANSIT-BUS DUTY PROFILE

Then compare EMS strategies using the same duty, component limits and initial states. Account for differences in final battery energy.

Model scope: prescribed speed; constant efficiencies; no detailed stack electrochemistry, voltage dynamics, thermal model or auxiliary loads.